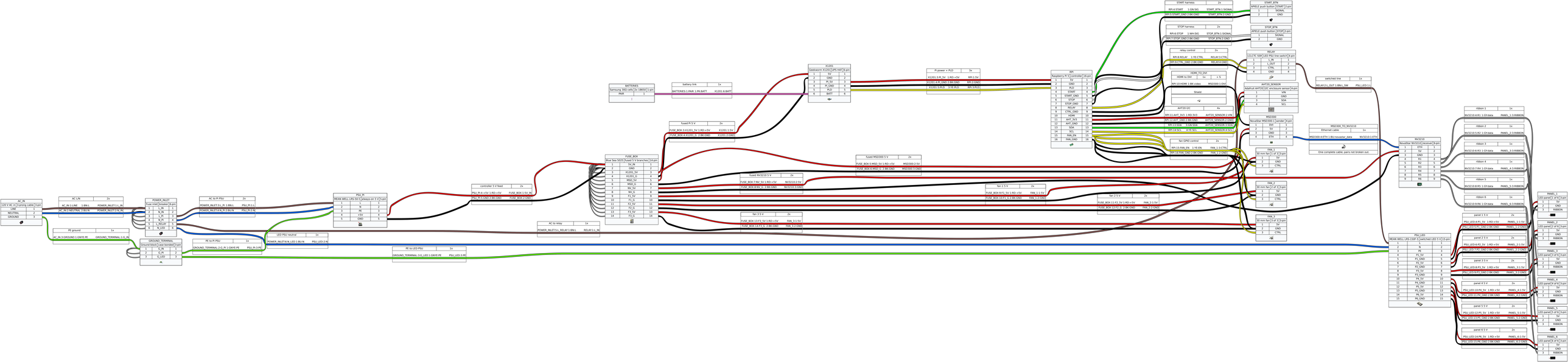

AC_IN

120 V AC in	3-prong cable	3-pin
LINE	1	
NEUTRAL	2	
GROUND	3	

AC L/N		2x
AC_IN:1:LINE	1:BN:L	POWER_INLET:1:L_IN
AC_IN:2:NEUTRAL	2:BU:N	POWER_INLET:2:N_IN

PE ground		1x
AC_IN:3:GROUND	1:GNYE:PE	GROUND_TERMINAL:1:G_IN

POWER_INLET

Fuse inlet	breaker	6-pin
1	L_IN	1
2	N_IN	2
3	L_PI	3
4	N_PI	4
5	L_RELAY	5
6	N_LED	6

GROUND_TERMINAL

Ground block	case bonded	3-pin
1	G_IN	1
2	G_PI	2
3	G_LED	3

AC to Pi PSU		2x
POWER_INLET:3:L_PI	1:BN:L	PSU_PI:1:L
POWER_INLET:4:N_PI	2:BU:N	PSU_PI:2:N

PE to Pi PSU		1x
GROUND_TERMINAL:2:G_PI	1:GNYE:PE	PSU_PI:3:PE

LED PSU neutral		1x
POWER_INLET:6:N_LED	1:BU:N	PSU_LED:2:N

PSU_PI

MEAN WELL LRS-50-5	always-on 5 V	5-pin
1	L	1
2	N	2
3	PE	3
4	+5V	4
5	GND	5

PE to LED PSU		1x
GROUND_TERMINAL:3:G_LED	1:GNYE:PE	PSU_LED:3:PE

controller 5 V feed		2x
PSU_PI:4:+5V	1:RD:+5V	FUSE_BOX:1:5V_IN
PSU_PI:5:GND	2:BK:GND	FUSE_BOX:2:GND

AC to relay		1x
POWER_INLET:5:L_RELAY	1:BN:L	RELAY:1:L_IN

BATTERIES

Samsung 30Q cells	2x 18650	1-pin
PAIR	1	

battery link		1x
BATTERIES:1:PAIR	1:PK:BATT	X1201:6:BATT

FUSE_BOX

Blue Sea 5025	fused 5 V branches	14-pin
1	5V_IN	1
2	GND	2
3	X1201_5V	3
4	X1201_G	4
5	MSD_5V	5
6	MSD_G	6
7	NV_5V	7
8	NV_G	8
9	F1_5V	9
10	F1_G	10
11	F2_5V	11
12	F2_G	12
13	F3_5V	13
14	F3_G	14

fused Pi 5 V		2x
FUSE_BOX:3:X1201_5V	1:RD:+5V	X1201:1:5V
FUSE_BOX:4:X1201_G	2:BK:GND	X1201:2:GND

X1201

Geekworm X1201	UPS HAT	6-pin
1	5V	1
2	GND	2
3	PI_5V	3
4	PI_GND	4
5	PLD	5
6	BATT	6

fused MSD300 5 V		2x
FUSE_BOX:5:MSD_5V	1:RD:+5V	MSD300:2:5V
FUSE_BOX:6:MSD_G	2:BK:GND	MSD300:3:GND

fused NV3210 5 V		2x
FUSE_BOX:7:NV_5V	1:RD:+5V	NV3210:2:5V
FUSE_BOX:8:NV_G	2:BK:GND	NV3210:3:GND

fan 3 5 V		2x
FUSE_BOX:13:F3_5V	1:RD:+5V	FAN_3:1:5V
FUSE_BOX:14:F3_G	2:BK:GND	FAN_3:2:GND

Pi power + PLD		3x
X1201:3:PI_5V	1:RD:+5V	RPI:1:5V
X1201:4:PI_GND	2:BK:GND	RPI:2:GND
X1201:5:PLD	3:YE:PLD	RPI:3:PLD

fan 1 5 V		2x
FUSE_BOX:9:F1_5V	1:RD:+5V	FAN_1:1:5V
FUSE_BOX:10:F1_G	2:BK:GND	FAN_1:2:GND

RPI

Raspberry Pi 5	controller	16-pin
1	5V	1
2	GND	2
3	PLD	3
4	START	4
5	START_GND	5
6	STOP	6
7	STOP_GND	7
8	RELAY	8
9	CTRL_GND	9
10	HDMI	10
11	AHT_3V3	11
12	AHT_GND	12
13	SDA	13
14	SCL	14
15	FAN_EN	15
16	FAN_GND	16

fan 2 5 V		2x
FUSE_BOX:11:F2_5V	1:RD:+5V	FAN_2:1:5V
FUSE_BOX:12:F2_G	2:BK:GND	FAN_2:2:GND

START harness		2x
RPI:4:START	1:GN:SIG	START_BTN:1:SIGNAL
RPI:5:START_GND	2:BK:GND	START_BTN:2:GND

STOP harness		2x
RPI:6:STOP	1:WH:SIG	STOP_BTN:1:SIGNAL
RPI:7:STOP_GND	2:BK:GND	STOP_BTN:2:GND

relay control		2x
RPI:8:RELAY	1:YE:CTRL	RELAY:3:CTRL
RPI:9:CTRL_GND	2:BK:GND	RELAY:4:GND

HDMI_TO_DVI

HDMI to DVI	1x	+ S
RPI:10:HDMI	1:BK:video	MSD300:1:DVI
	Shield	

AHT20 I2C		4x
RPI:11:AHT_3V3	1:RD:3V3	AHT20_SENSOR:1:VIN
RPI:12:AHT_GND	2:BK:GND	AHT20_SENSOR:2:GND
RPI:13:SDA	3:GN:SDA	AHT20_SENSOR:3:SDA
RPI:14:SCL	4:YE:SCL	AHT20_SENSOR:4:SCL

fan GPIO control		2x
RPI:15:FAN_EN	1:YE:EN	FAN_1:3:CTRL
RPI:16:FAN_GND	2:BK:GND	FAN_1:2:GND

START_BTN

APIELE push button	START	2-pin
1	SIGNAL	
2	GND	

STOP_BTN

APIELE push button	STOP	2-pin
1	SIGNAL	
2	GND	

RELAY

LCLCTC SSR	LED PSU line switch	4-pin
1	L_IN	1
2	L_OUT	2
3	CTRL	3
4	GND	4

AHT20_SENSOR

Adafruit AHT20	I2C enclosure sensor	4-pin
1	VIN	
2	GND	
3	SDA	
4	SCL	

MSD300

NovaStar MSD300-1	sender	4-pin
1	DVI	1
2	5V	2
3	GND	3
4	ETH	4

FAN_1

30 mm fan	1 of 3	3-pin
1	5V	
2	GND	
3	CTRL	

FAN_2

30 mm fan	2 of 3	3-pin
1	5V	
2	GND	
3	CTRL	

FAN_3

30 mm fan	3 of 3	3-pin
1	5V	
2	GND	
3	CTRL	

switched line		1x
RELAY:2:L_OUT	1:BN:L_SW	PSU_LED:1:L

MSD300_TO_NV3210

Ethernet cable		1x
MSD300:4:ETH	1:BU:novastar_data	NV3210:1:ETH

One complete cable; pairs not broken out.

NV3210

NovaStar NV3210	receiver	9-pin
1	ETH	1
2	5V	2
3	GND	3
4	R1	4
5	R2	5
6	R3	6
7	R4	7
8	R5	8
9	R6	9

PSU_LED

MEAN WELL LRS-150F-5	switched LED 5 V	15-pin
1	L	1
2	N	2
3	PE	3
4	P1_5V	4
5	P1_GND	5
6	P2_5V	6
7	P2_GND	7
8	P3_5V	8
9	P3_GND	9
10	P4_5V	10
11	P4_GND	11
12	P5_5V	12
13	P5_GND	13
14	P6_5V	14
15	P6_GND	15

ribbon 1		1x
NV3210:4:R1	1:GY:data	PANEL_1:3:RIBBON

ribbon 2		1x
NV3210:5:R2	1:GY:data	PANEL_2:3:RIBBON

ribbon 3		1x
NV3210:6:R3	1:GY:data	PANEL_3:3:RIBBON

ribbon 4		1x
NV3210:7:R4	1:GY:data	PANEL_4:3:RIBBON

ribbon 5		1x
NV3210:8:R5	1:GY:data	PANEL_5:3:RIBBON

ribbon 6		1x
NV3210:9:R6	1:GY:data	PANEL_6:3:RIBBON

panel 1 5 V		2x
PSU_LED:4:P1_5V	1:RD:+5V	PANEL_1:1:5V
PSU_LED:5:P1_GND	2:BK:GND	PANEL_1:2:GND

panel 2 5 V		2x
PSU_LED:6:P2_5V	1:RD:+5V	PANEL_2:1:5V
PSU_LED:7:P2_GND	2:BK:GND	PANEL_2:2:GND

panel 3 5 V		2x
PSU_LED:8:P3_5V	1:RD:+5V	PANEL_3:1:5V
PSU_LED:9:P3_GND	2:BK:GND	PANEL_3:2:GND

panel 4 5 V		2x
PSU_LED:10:P4_5V	1:RD:+5V	PANEL_4:1:5V
PSU_LED:11:P4_GND	2:BK:GND	PANEL_4:2:GND

panel 5 5 V		2x
PSU_LED:12:P5_5V	1:RD:+5V	PANEL_5:1:5V
PSU_LED:13:P5_GND	2:BK:GND	PANEL_5:2:GND

panel 6 5 V		2x
PSU_LED:14:P6_5V	1:RD:+5V	PANEL_6:1:5V
PSU_LED:15:P6_GND	2:BK:GND	PANEL_6:2:GND

PANEL_1

LED panel	1 of 6	3-pin
1	5V	
2	GND	
3	RIBBON	

PANEL_2

LED panel	2 of 6	3-pin
1	5V	
2	GND	
3	RIBBON	

PANEL_3

LED panel	3 of 6	3-pin
1	5V	
2	GND	
3	RIBBON	

PANEL_4

LED panel	4 of 6	3-pin
1	5V	
2	GND	
3	RIBBON	

PANEL_5

LED panel	5 of 6	3-pin
1	5V	
2	GND	
3	RIBBON	

PANEL_6

LED panel	6 of 6	3-pin
1	5V	
2	GND	
3	RIBBON	